

RESEARCH ARTICLE OPEN ACCESS

High-Performance Zero-Gap Glycerol-Fed Electrolyzer for C₃ Chemicals and Hydrogen Production

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Received: 7 August 2025 | **Revised:** 24 November 2025 | **Accepted:** 11 December 2025

Keywords: biomass conversion | electrocatalyst | glycerol | hydrogen | platinum

ABSTRACT

The electrochemical oxidation of biomass-derived glycerol offers a promising low-voltage alternative to water oxidation in electrolyzers, enabling the co-production of hydrogen and value-added chemicals. However, achieving high conversion rates at current densities above 300 mA cm⁻² remains challenging due to the rapid deactivation of platinum-based catalysts. Here, we present a membrane electrode assembly (MEA) featuring a platinum-decorated nickel foam (Pt/NiF) anode that sustains operation for 24 h at 500 mA cm⁻² with an average cell voltage of just 1.21 V, outperforming all previously reported glycerol-fed electrolyzers operating below 1.5 V. The system exhibits >88% selectivity toward C₃ products, achieving 227 mA cm⁻² partial current density for lactic acid and 9% single-pass glycerol conversion. In situ impedance spectroscopy identifies voltage-dependent regimes linked to platinum hydroxide formation, glycerol oxidation, and oxygen evolution. Systematic variation of electrolyte composition and temperature reveals an optimized window (1.2–1.4 V, 55–65°C) for sustained performance. Under these conditions, a single 24 h cycle co-generates ~175 mmol of H₂ and 45 mmol of C₃ products. These results establish new operational and mechanistic benchmarks for efficient, low-voltage electrochemical valorization of biomass-derived polyols at industrially relevant rates.

1 | Introduction

As the environmental impacts of fossil fuel consumption, including rising greenhouse gas emissions and accelerating climate change, continue to grow, the global demand for carbon-neutral energy sources has become urgent. Among the available alternatives, hydrogen (H₂) stands out for its high gravimetric energy density (142 MJ kg⁻¹), flexibility as an energy carrier, and potential for carbon-free combustion [1]. Yet, nearly 98% of global H₂ production, projected to reach 100 million tonnes by 2025, still relies on fossil fuel-based processes like steam methane reforming and coal gasification, emitting ~920 Mt of CO₂ annually [1, 2].

To mitigate these emissions associated with H₂ production, water electrolysis is recognized as a sustainable pathway for generating “green H₂” using water and renewable electricity sources such as wind, solar, and hydropower [3, 4]. However, the widespread adoption of water electrolyzer technologies remains limited due to their high energy requirements, primarily due to the sluggish kinetics and significant overpotential associated with the oxygen evolution reaction (OER) at the anode, which accounts for approximately 90% of the electrical energy consumption in water electrolysis [5–7]. This energy requirement and kinetic bottleneck limits the economic viability of large-scale H₂ production via electrolysis.

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In addition to efforts focused on developing more active and durable anode catalysts for the OER, an alternative and increasingly attractive strategy involves replacing the OER at the anode of water electrolyzers with thermodynamically more favorable oxidation reactions, such as the glycerol oxidation reaction (GOR). Glycerol, a major waste product from the biofuel, biodiesel, and soap industries, is produced in vast quantities globally, with the biodiesel industry alone generating approximately 3–4 million tons annually [8, 9]. Transforming glycerol from a waste product ($\sim \$0.1\text{--}0.3 \text{ USD kg}^{-1}$ for 80% purity) into valuable chemicals through selective electrooxidation not only reduces the energy requirements of electrolyzers but also supports industrial circularity [10]. For example, oxidation of glycerol can yield economically valuable compounds such as glyceric acid, lactic acid, and glycolic acid, some of which are valued up to 1000 times higher than the feedstock [11]. Moreover, GOR has a standard reversible potential of a $0.003 \text{ V}_{\text{SHE}}$, nearly 1.2 V less than that of OER. Therefore, substituting GOR for OER in water electrolyzers offers a compelling dual advantage: reducing the energy input requirement for H_2 production via electrolysis and promoting waste valorization for achieving a circular economy. However, the implementation of GOR in practical electrolyzer systems remains technically challenging due to the complex multi-pathway nature of the reaction and the difficulty in achieving single product selectivity and long-term catalyst stability [12, 13].

As a result, recent research has increasingly focused on the development of electrocatalysts that exhibit high activity, operational durability, and tunable selectivity under working conditions toward GOR. Yet, despite substantial progress in catalyst engineering and mechanistic understanding, only a limited number of studies have successfully demonstrated the integration of GOR with H_2 production at current densities $>300 \text{ mA cm}^{-2}$ [14–17]. Initial investigations by Chen et al. demonstrated substantial improvements in the electrical energy efficiency using a Pd-based catalyst to facilitate GOR, achieving high current densities of 300 mA cm^{-2} at cell voltages below 1.1 V, significantly lower than the cell voltages of commercial water electrolyzers employing OER (typically between 1.8 and 2.2 V) [18]. Subsequent studies, such as those by Yan et al., further emphasized the potential of using cost-effective nickel foam (NiF) substrates as a GOR anode by demonstrating that $\text{Au}_3\text{Ag}_1/\text{NiF}$ catalysts could achieve current densities above 1000 mA cm^{-2} in a three-electrode cell at potentials around 1.2 VRHE through linear sweep voltammetry (LSV) measurements [19]. However, these current densities could not be translated to a flow electrolyzer during long-term chronoamperometry tests, whereby, despite high selectivity toward lactic acid (up to 70%), the current density dropped to roughly 50 mA cm^{-2} (at 1.4 V) [20], highlighting the need for understanding and tuning electrolyzer operation to achieve sustained high current densities. In addition to hydrogen evolution reaction (HER), pairing GOR with other cathodic reactions, such as electrochemical reduction of carbon dioxide [7, 21] or carbon mono-oxide [22], as well as nitrate-to-ammonia conversions [23, 24] has also gained popularity in the last few years, demonstrating the broad practicality of organic oxidation and its potential to replace the energy-intensive OER. Yadegari and coworkers provide an excellent example of this by pairing carbon monoxide reduction with GOR, sustaining a current density of 180 mA cm^{-2} at a cell voltage of $\sim 1.34 \text{ V}$ with 75% Faradaic efficiency toward C3 products at the anode and 72%

to C_2 chemicals at cathode [22]. Despite progress in electrolyzer design and catalyst development, a comprehensive understanding of the GOR under realistic membrane electrode assembly (MEA) operating conditions remains lacking. In particular, the influence of key operational parameters, such as cell voltage, temperature, glycerol concentration, and electrolyte composition, on catalytic activity, product selectivity, and system stability has not been thoroughly characterized. Moreover, the complex interplay of electrochemical and non-Faradaic pathways in GOR [25], combined with the competing OER that can occur in the MEA under high voltage operation, often leads to inconsistent or misinterpreted performance metrics across the literature.

In this work, we systematically investigate the effects of key MEA operating parameters using platinum nano-dendrites supported on nickel foam (Pt/NiF) as a benchmark anode electrode for the GOR. Our research elucidates the underlying mechanisms of GOR on Pt and Ni sites, resolving some of existing inconsistencies in the literature regarding the role of Pt–OH in the oxidation of glycerol. By applying the findings in this work to design a high conversion MEA reactor, our system exhibits current densities exceeding 500 mA cm^{-2} and $>88\%$ selectivity toward high-value C_3 products at an operating voltage of just 1.21 V, thereby outperforming all previously reported GOR-based MEAs coupled with H_2 production. Furthermore, long-term dynamic operation at 500 mA cm^{-2} reveals a stable cell voltage of $<1.25 \text{ V}$, maintaining nearly 50% selectivity toward lactic acid and achieving nearly 9% single-pass glycerol conversion for over 24 h. Through a series of systematic voltage-dependent experiments, we also introduce two distinct operating regimes for glycerol-fed MEAs: i) an OER-free GOR zone, and ii) a GOR/OER hybrid zone. Our research enables achieving high conversion and selectivity within the OER-free zone instead of the GOR/OER hybrid zone despite lower current densities, moving toward low-energy-input H_2 production units. These findings provide mechanistic insight into GOR under practical electrolysis conditions and establish a framework for the design of dual-function electrolyzers that couple low-voltage H_2 production with biomass valorization.

2 | Results and Discussion

2.1 | Electrochemical Performance of Electrodes in 3-Electrode H-Cell

To assess the electrochemical behavior of the synthesized Pt/NiF electrodes in the absence of glycerol, LSV measurements were conducted in 3M KOH at a scan rate of 10 mV s^{-1} (see Methods for synthesis details). As shown in Figure S1, the Pt/NiF electrode exhibits a sharp increase in current density at potentials around $1.55 \text{ V}_{\text{RHE}}$, indicative of the onset for the OER. When compared to a commercial IrO_x electrode, Pt/NiF displays a noticeably higher onset potential, while the measured current density demonstrates noise at current densities exceeding 200 mA cm^{-2} , likely due to bubble formation and detachment processes on the surface and within the porous structure of the NiF, which can impact mass transport. In contrast, comparing Pt/NiF with a bare NiF (see Figure S2A–F for comparison of surface morphologies) reveals that the unmodified NiF substrate has an even higher overpotential for OER, with Tafel slopes comparison shown in Figure S3. We believe the difference between the OER activity

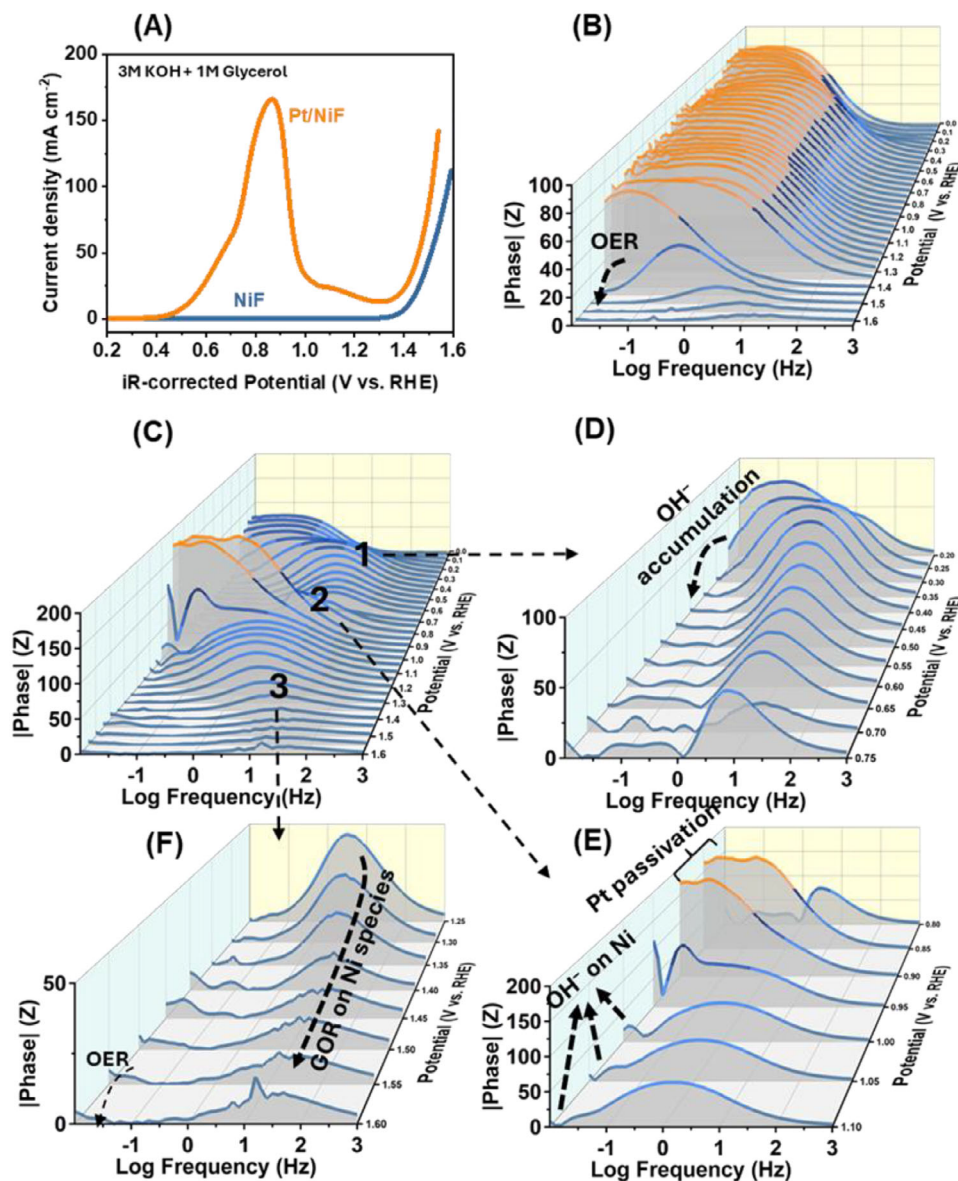


FIGURE 1 | Electrochemical behavior of Pt/NiF and NiF electrodes in glycerol-containing alkaline electrolyte. (A) LSV profiles of Pt/NiF and bare NiF in 3 M KOH containing 1 M glycerol, recorded at a scan rate of 10 mV s^{-1} . (B) Bode phase plots from in situ electrochemical impedance spectroscopy (EIS) of Pt/NiF in 3 M KOH at various applied potentials (0–1.6 VRHE). (C) Bode phase plots of Pt/NiF in 3 M KOH + 1 M glycerol across the same potential range. (D–F) Zoomed-in views of the EIS data in panel C, highlighting three distinct potential zones: (D) Zone 1 (0.2–0.75 VRHE), (E) Zone 2 (0.8–1.1 VRHE), and (F) Zone 3 (1.25–1.6 VRHE).

of NiF and Pt/NiF is likely attributed to the higher density of Ni oxide species on Pt/NiF compared to NiF, particularly β -NiOOH, formed during the galvanic displacement process in HClO_4 . Our previous work [26] confirmed the formation of β -NiOOH through X-ray photoelectron spectroscopy (XPS) analyses of the Ni 2p region on Pt/NiF surfaces. Species widely recognized with their catalytically active phases for OER in alkaline media that can initiate the OH^- adsorption at fairly low overpotentials (about $1.45 \text{ V}_{\text{RHE}}$) [27, 28] (see Section S1 and Equations S1–S5 for the detailed alkaline OER reaction mechanisms on Ni-based electrode).

Upon the introduction of 1 M glycerol into the 3 M KOH electrolyte, the Pt/NiF electrode demonstrates noticeably different electrochemical behavior compared to when glycerol is absent

(Figure 1A). A sharp oxidation peak appears centered at $\sim 0.86 \text{ V}$ versus RHE, characteristic of the adsorption and oxidation of glycerol on Pt-based catalysts [29]. However, increasing the potential beyond $\sim 0.9 \text{ V}_{\text{RHE}}$ results in a significant drop in current density. This decline is commonly attributed to the formation of surface Pt-(hydr)oxide species, which passivate the surface of the catalytically active sites and inhibit catalytic turnover [30]. Additionally, surface poisoning by strongly adsorbed carbonaceous intermediates, particularly $\ast\text{CO}$, has been implicated in activity loss [31]. When the applied potential exceeds $\sim 1.3 \text{ V}_{\text{RHE}}$, a second substantial increase in current density is observed, indicating reactivation of the catalyst. This second onset of catalytic activity is likely due to the onset of additional glycerol oxidation pathways (possibly on Ni) and the concurrent contribution of the OER. As such, at potentials above $1.4\text{--}1.5 \text{ V}_{\text{RHE}}$, glycerol oxidation and

OER occur as competing reactions, each influencing the overall electrochemical performance of the Pt/NiF electrode. The LSV profile of bare NiF, shown in Figure 1A, also exhibits a notable increase in current density at approximately 1.3 V versus RHE, which is significantly lower than the onset potential of the OER observed in (Figure S1). The origin of glycerol oxidation on Ni nanostructures is currently debatable, but as we previously reported [26], NiF can catalyze glycerol oxidation via an “indirect charge transfer” mechanism, wherein β -NiOOH is electrochemically reduced to β -Ni(OH)₂ while concurrently oxidizing glycerol to various products. This mechanism relies critically on the reversible redox transition between Ni(OH)₂ and NiOOH, which facilitates the adsorption of glycerol molecules through hydrogen bonding, followed by subsequent dehydrogenation and oxidation steps. Therefore, in situ measurements, such as X-ray absorption spectroscopy and vibrational spectroscopies (e.g., ATR-SEIRAS or in situ Raman) under relevant operating conditions, would be powerful tools to directly probe the evolution of Pt–OH and Ni–O species and further refine the mechanistic assignments proposed here.

Knowing that the Pt/NiF catalyst can oxidize glycerol both within an OER-free potential range ($<0.9 V_{\text{RHE}}$) and a mixed GOR/OER range ($>1.5 V_{\text{RHE}}$), it is critical to distinguish between these two electrochemical zones prior to integrating the catalyst into a MEA. To this end, we performed a series of potential-dependent in situ EIS measurements to probe the catalytic kinetics and interfacial characteristics of the Pt/NiF electrode associated with the OER and GOR under working conditions (Figure 1B,C). In the Bode plots, there are usually two distinct frequency regions that are important: a low-frequency region (10^{-2} – 10^0 Hz), typically governed by mass transport and surface transformations, and a mid to high-frequency region (10^0 – 10^3), associated with charge transfer dynamics [32]. Figure 1B shows the Bode response of Pt/NiF in 3 M KOH (without glycerol) at applied potentials ranging from 0 to $1.6 V_{\text{RHE}}$. Up to approximately $1.4 V_{\text{RHE}}$, the phase angle remains close to 90° in the low-frequency range, indicating capacitive behavior and/or electrochemical inactivity of the catalyst for OER (see Figure S4A,B for redox features of pure Pt and NiF in 3M KOH). However, beyond $1.4 V_{\text{RHE}}$, the plot shows decreased peak intensity in the ultralow-frequency region (below $<10^{-2}$). This transition is indicative of OH[−] accumulation and structural changes on the catalyst surface, likely associated with the formation of oxidized species and the initiation of OER activity as reported previously [33]. The appearance of these features in the Bode plot, particularly the shift of impedance peaks toward the mid-frequency range, further supports the onset of OER at potentials above $\sim 1.4 V_{\text{RHE}}$.

For Pt/NiF in the presence of glycerol, the phase angle behavior as a function of applied potential becomes more complex (Figure 1C). To facilitate interpretation, we have categorized Figure 1C into three distinct potential ranges: Zone 1 (0.2 to $0.75 V_{\text{RHE}}$), Zone 2 (0.85 to $1.1 V_{\text{RHE}}$), and Zone 3 (1.25 to $1.6 V_{\text{RHE}}$). In Zone 1 (Figure 1D), the low-frequency phase response begins near 90° , indicative of capacitive behavior and a largely inactive surface. However, as the potential increases to $\sim 0.3 V_{\text{RHE}}$, a sharp decline in phase angle is observed. This transition, as previously claimed, likely corresponds to the initial accumulation of *OH species on the surface of Pt [34]. And marks the electrochemical activation of the catalyst toward GOR, supporting the hypothesis

that Pt–OH species can initiate the alcohol oxidation at potential range of 0.3 – $0.4 V_{\text{RHE}}$ on the surface of Pt. This interpretation is further supported by the LSV profile in Figure 1A, where a sharp rise in current is observed over the same potential window.

As the potential increases beyond $\sim 0.45 V_{\text{RHE}}$, the Bode phase peaks shift toward the mid-frequency range (10^0 – 10^2 Hz), while the overall phase angle decreases. This shift reflects a change in the dominant interfacial processes: the surface becomes increasingly populated with reactive *OH species, and the system transitions from double-layer charging to a regime dominated by Faradaic charge transfer. These kinetic features are consistent with the direct electrooxidation of glycerol and correlate well with the increased current densities observed in the 0.4 – $0.7 V_{\text{RHE}}$ range of the CV profile (Figure S5), further confirming the catalytic activity of Pt/NiF under these conditions. Comparison of the CV profile of Pure Pt and Pt/NiF in the presence of glycerol (Figure S5) also reveals lower onset potentials for Pt/NiF, and a maximum at more oxidative potentials, indicating the presence of Ni delays Pt deactivation.

Comparing the Bode plots at $0.7 V_{\text{RHE}}$ and $0.75 V_{\text{RHE}}$ (Figure 1E) shows that the mid-frequency phase response exhibits a considerable rise, indicating that glycerol oxidation is becoming more sluggish on the Pt/NiF with an increase in electrode potential. This trend is likely due to catalyst deactivation, either through surface passivation by Pt-(hydr)oxide or poisoning by adsorbed intermediates. Further supporting this hypothesis is the emergence of irregular features in the Bode plots within the 0.85 – $0.95 V_{\text{RHE}}$ range, including signal instabilities and phase angles exceeding 90° , which are commonly associated with catalyst degradation and non-ideal interfacial behavior. In other words, under alkaline conditions, these Pt-(hydr)oxide species can oxidize glycerol, and the resulting intermediates can possibly form a layer that remains adsorbed on the surface and passivates Pt. As the potential approaches $0.85 V_{\text{RHE}}$, the phase angle increases sharply, indicating extensive surface oxidation and deactivation of Pt (due to the lower conductivity of Pt–O_x, the Bode plot shows abnormal shifts in phase angle ($>90^\circ$)), along with the concurrent inactivity of Ni sites. However, upon further increasing the potential, Ni–OH species begin to form and become electrochemically active, as reflected by a rise in the low- to mid-frequency phase angles.

We believe this potential range is correlated with the rapid decline in the GOR current density in the LSV graph shown in Figure 1A. This phenomenon can also be seen during the chronoamperometry tests of Pt/NiF, in the presence of 1M glycerol/3M KOH where applying potentials between 0.85 and 0.95 enables a high current density to start; however, it is followed by a rapid decline, possibly due to surface poisoning by CO intermediate and formation of Pt–O_x layer (Figure S6). However, such declines are observed considerably higher in pure Pt compared to Pt/NiF, indicating the role of Ni in poison tolerance of Pt, as discussed elsewhere. However, beyond $0.95 V_{\text{RHE}}$, two distinct responses emerge in the Bode plots: a low-frequency feature ($\sim 10^{-2}$ Hz), attributed to OH[−] accumulation and surface restructuring, and a mid-frequency (10^{-1} – 10^1 Hz), likely corresponding to glycerol oxidation on the underlying Ni substrate following partial passivation of the Pt surface (Figure 1E) [35]. These interpretations align well with the

LSV behavior of bare NiF (Figure 1A), where a gradual rise in current is observed beginning around 1.2 V_{RHE} (Figure 1A).

As the applied potential increases further, the mid-frequency phase peak steadily decreases up to ~1.55 V_{RHE} (Figure 1F), indicating accelerated reaction kinetics for Ni-mediated glycerol oxidation. At approximately 1.55 V_{RHE}, however, a new low-frequency peak emerges, marking the re-accumulation of OH[−] species and signaling the onset of OER [32]. It is important to note that EIS measurements become increasingly unreliable at potentials above 1.6 V due to system instability and gas bubble formation (O₂), which disrupts signal integrity. Consequently, we did not observe the complete disappearance of the low-frequency response, which would otherwise help define the precise OER onset in the presence of glycerol. Nonetheless, comparison of the OER onset in the absence (Figure 1B) and presence (Figure 1F) of glycerol suggests that glycerol oxidation delays OER initiation due to OH[−] consumption, thereby requiring a higher applied potential to sustain OER.

Another critical consideration when transitioning from fundamental electrochemical characterization to practical MEA operation is the mismatch between cell voltages measured in 2-electrode configurations and the reaction potentials determined in 3-electrode systems. In a 2-electrode cell, the individual anode and cathode potentials are not referenced against a stable reference electrode (e.g., RHE) but rather to each other, typically with the cathode serving as the implicit reference, particularly when the cathodic reaction is the HER on Pt. To bridge this gap and roughly estimate the actual electrode potentials in our MEA, while acknowledging that transport properties and local reaction environments in MEA can influence these values, we conducted iR-corrected LSV measurements for both the anode (Pt/NiF) and cathode (Pt/C) in their respective half-cell electrolytes.

As shown in Figure S7, in the absence of glycerol (3 M KOH), achieving a current density of 100 mA cm^{−2} requires an electrode potential difference of approximately 1.8 V_{RHE}, corresponding to anode potential of 1.58 V_{RHE} (Pt/NiF) and cathode potential of −0.22 V_{RHE} (Pt/C). However, when the same measurement is conducted in the presence of 1 M glycerol, the required potential difference sharply decreases to ~0.98 V to achieve the same current density. Under these conditions, the Pt/NiF anode operates at 0.76 V_{RHE}, while the Pt/C cathode remains at −0.22 V_{RHE}. This shift of 800 mV in the anodic potential to achieve 100 mA cm^{−2} of current density toward the GOR versus the OER highlights the lower energy input that is needed when incorporating the GOR at the anode. These findings emphasize the potential of organic electrolysis, particularly glycerol oxidation, to significantly reduce the energy input required for H₂ production, an important consideration for the industrial viability of next-generation electrolyzers.

2.2 | Benchmarked Operation of MEA at Room Temperature

To evaluate the electrochemical performance of the Pt/NiF anode in an industrially relevant flow-type reactor, a 2-electrode MEA was tested with continuous flow of both a 3 M KOH + 1 M glycerol anolyte and 3 M KOH catholyte (Figure 2A,B). Figure 2C

shows the LSVs conducted at 10 mV s^{−1} in both the presence (anolyte: 3 M KOH + 1 M glycerol) and absence (anolyte: 3 M KOH) of glycerol (see Methods for more details). In the absence of glycerol in the anolyte (blue curve), a characteristic oxidative peak associated with Ni(OH)₂ to NiOOH is observed at ~1.5 V, followed by a steady increase in the current density due to the OER. The noise in the measured current density observed beyond 2 V can be attributed to vigorous O₂ bubble generation at the anode, which impacts mass transport. Conversely, when 1M glycerol is present in the anolyte (orange curve), the current density first increases considerably up to ~1.4 V, indicative of the GOR occurring on the Pt catalyst. Then, similar to the trend observed in the CV profile of the Pt/NiF electrode (Figure S5), the current density drops significantly when the cell voltage increases further, likely due to surface passivation of Pt sites by Pt (hydr)oxide and/or poisoning by strongly adsorbed intermediates such as *CO [36, 37]. Later, at voltages beyond 1.75 V, the current density rises again, suggesting the reactivation of catalytic sites and the simultaneous onset of Ni-mediated oxidation and OER processes. These results reveal a distinct dual-zone behavior in MEA operation: at cell voltages <1.5 V, the anodic current is primarily derived from glycerol oxidation (GOR zone), while at voltages >1.5 V, both GOR and OER processes contribute (GOR/OER zone), similar behavior to what was observed in the 3-electrode H-cell measurements.

Motivated by the distinct operational zones identified earlier, we further explored how electrolyte composition influences the performance of the MEA, particularly in maximizing GOR activity in the GOR zone. To this end, we systematically varied the glycerol concentration and, by extension, the glycerol to KOH ratio in the anolyte, as well as the KOH concentration in the catholyte, conducting LSVs under each condition. The objective was to identify the operating parameters that yield the highest GOR current density in the GOR zone. As shown in Figure 2D, increasing the glycerol concentration from 0 to 1.2 M in 3 M KOH results in a proportional increase in the current density toward the GOR, attributed to the improved mass transport leading to a greater availability of reactant molecules at the catalyst surface (see Figures S8 and S9A–C for uncompensated cell voltage and EIS graphs in Figure 2C,D, respectively). However, further increasing the glycerol concentration beyond 1.2 M does not lead to additional improvements and, in some cases (e.g., beyond 2 M), leads to performance deterioration, likely due to mass transport limitations associated with the high viscosity of glycerol (~1500 mPa·s at 25°C) [38].

In a separate experiment, the glycerol concentration in the anolyte was fixed at 1.2 M (identified earlier as the value that provides the highest current density toward the GOR) while the KOH concentration in the catholyte was varied to examine the effect on MEA performance. As shown in Figures S10 and S11, increasing the KOH concentration in the catholyte from 0 to 6 M leads to a marked enhancement in current density within the GOR zone, while concurrently resulting in a decreased current density in the GOR/OER zone. These results can be explained by considering two interrelated phenomena in the MEA system: i) the influence of KOH concentration on the hydrogen evolution reaction (HER) kinetics at the Pt/C cathode, and ii) the role of OH[−] transport from the catholyte to the anode, which affects the local reaction environment for glycerol oxidation.

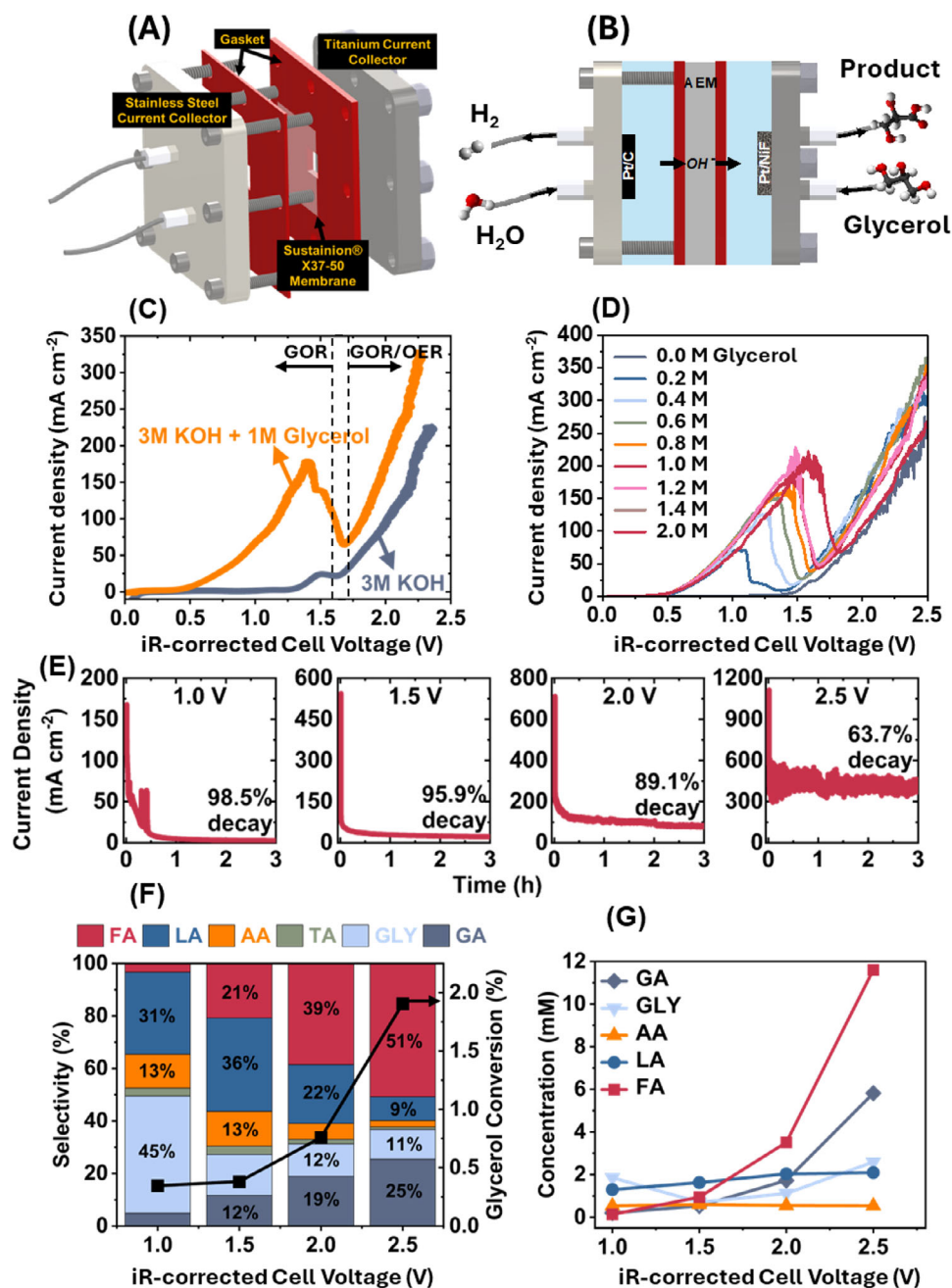


FIGURE 2 | Electrochemical performance and product distribution in MEA during glycerol oxidation. A,B) Schematic illustrations of the membrane electrode assembly configuration used in this study. (C) LSV profiles of the MEA in the presence and absence of 1 M glycerol in the anolyte, recorded at a scan rate of 10 mV s^{-1} (anolyte: 3 M KOH; catholyte: 3 M KOH), indicating two voltage zones of GOR and GOR/OER (D) LSV profiles of the MEA with varying glycerol concentrations in the 3 M KOH anolyte, recorded at 10 mV s^{-1} . (E) Chronoamperometry measurements of the MEA at cell voltages ranging from 1.0 to 2.5 V, highlighting current density decay over time. (F) Bar graph showing product selectivity across different cell voltages after 3 h chronoamperometry, along with corresponding single-pass glycerol conversion efficiencies (FA=formic acid, LA=lactic acid, AA=acetic acid, GLY=glyceric acid, GA=glycolic acid). (G) Variation in GOR product concentrations during chronoamperometry at different cell voltages.

The HER LSV profiles of Pt/C as a function of catholyte KOH concentration are shown in Figure S12. The electrode potentials have been iR-corrected using solution resistance values obtained from EIS measurements under steady-state conditions (see Figure S13 for the change in the solution resistance as a function of KOH concentration). This correction accounts for variations in bulk ionic conductivity, allowing for a more accurate comparison of intrinsic catalytic behavior. According to the LSV profiles, two distinct regimes emerge. i) Increasing the KOH

concentration from 0.1 to 4 M leads to an improvement in HER activity. Since ohmic losses have already been compensated for, this enhancement likely originates from changes at the electrode-electrolyte interface. Prior studies have suggested that moderate alkalinity can optimize interfacial water orientation and enhance the kinetics of water dissociation on Pt, improving proton availability for the Volmer step [39]. ii) Beyond 4 M, a decline in performance is observed. This trend aligns with previous work by Guha et al. [40, 41], which showed that

highly concentrated alkaline solutions (around 6 M) foster the formation of tetrahedrally coordinated water clusters near the Pt surface. These structured networks inhibit water dissociation and hinder proton delivery, thereby slowing HER kinetics. In fact, at lower alkalinity, HER proceeds primarily via the Heyrovsky step (electrochemical desorption), whereas at higher alkalinity, the Volmer step (proton adsorption/dissociation) becomes limiting due to the more structured interfacial environment and reduced proton availability [39].

With the optimal anolyte (1.2 M glycerol in 3 M KOH) and catholyte (4 M KOH) identified, we next performed chronoamperometry measurements on the MEA at various cell voltages, followed by quantitative analyses of the products formed via GOR. A key observation in the chronoamperometry measurements across all voltages (Figure 2E) was a rapid drop in current density shortly after the start of the experiment. At cell voltages below 2 V, the current density of the Pt/NiF electrode declined by more than 90% within minutes, whereas at voltages around 2.5 V, less decay was observed (68%). This contrast arises from the electrochemical zones previously discussed. Below 1.5 V, the current originates almost entirely from GOR, where Pt deactivation due to the formation of Pt-(hydr)oxide or surface poisoning species (e.g., $\ast\text{CO}$ and intermediates) causes severe activity loss. In contrast, above ~ 1.5 V, the OER contributes to the current density, stabilizing the current in the GOR/OER zone due to continuous O_2 evolution on the NiOOH surface within the electrode.

To evaluate the glycerol oxidation product obtained at different voltages, a series of 3 h chronoamperometry experiment were conducted at cell voltage between 0.5 and 2.5 V with a single-pass glycerol configuration (Figure 2F). At a cell voltage of 1.0 V, glyceric acid (a C_3 product) was the major oxidation product, being produced with 45% selectivity, accompanied by 31% GOR selectivity toward lactic acid, leading to a cumulative C_3 selectivity near 84%. As the cell voltage increases, C—C bond cleavage becomes more dominant, shifting the product distribution toward smaller molecules. At 1.5 V, cumulative C_3 selectivity drops to 55%, and further to just 21% at 2.0 V. In contrast, the selectivity toward formic acid, a C_1 product, increases sharply from 2% to $\sim 51\%$ as the cell voltage rises from 1.0 to 2.5 V. Moreover, the single-pass glycerol conversion increases from 0.5% (producing ~ 6 μmoles of product) to over 2% (approximately 25.5 moles) as the cell voltage increases from 1.0 to 2.5 V. These results present a fundamental operational dilemma in MEA-driven glycerol electrooxidation: should the system be operated at low voltages (<1.5 V) to prioritize the selective production of high-value C_3 chemicals such as glyceric and lactic acid, or at higher voltages (>1.5 V) to maximize overall glycerol conversion at the expense of producing lower-value products such as formic acid? This topic will be explored in the following sections.

2.3 | Impact of Temperature on Performance Metrics of Catalyst and MEA Operation

The impact of temperature on the performance of glycerol-fed MEAs has remained largely underexplored, primarily due to the complexity of interpreting changes in the GOR mechanism, particularly in voltage regions where GOR and the OER overlap.

To gain insight into how increasing temperature can impact the performance of Pt/Ni electrodes, the CV profiles of the Pt/NiF electrode in the presence of 1.2 M glycerol in 3 M KOH at temperatures ranging from 25 to 55°C were recorded (Figure 3A). Several key observations emerge from these temperature-dependent profiles: i) As the temperature increases, the onset potential for GOR (defined as the point where the oxidative current begins to rise significantly) shifts to lower potential. This shift is attributed to the formation of surface-bound $\text{Pt}-\text{OH}_x$ species that is enhanced at increased temperatures [42], which we previously have shown to contribute toward the catalytic glycerol oxidation (see Figure 1D). These species enhance glycerol adsorption and subsequent oxidation via both the carbon (direct $\text{Pt}-\ast\text{C}$ interaction) and oxygen terminals (hydrogen bonding via hydroxyl groups in glycerol); [43] ii) The potential at which the maximum current density is observed (peak potential) shifts substantially from 0.92 V_{RHE} at 25°C to 1.16 V_{RHE} at 55°C, indicating an extended operational window before catalyst poisoning/deactivation; iii) the peak current density increases from 167 to 625 mA cm^{-2} (a nearly fourfold enhancement), demonstrating the strong temperature dependence of GOR performance in this electrode, iv) the oxidation peak observed during the reverse scan also increases in magnitude with temperature (see Figure S14 for the comparison of forward and reverse oxidation peaks). The origin of this reverse oxidation peak has been an area of controversy in the last few years, largely associated with the oxidation of glycerol on Pt following the reductive removal of oxide species [29, 44], or oxidation of $\ast\text{CO}$ on Pt [45]. In either case, the enhanced reverse scan currents at higher temperatures indicate both increased catalytic performance and improved surface recovery and reactivation of the Pt/NiF electrode.

Building on this temperature-dependent GOR performance of the Pt/NiF electrode, it becomes essential to examine how glycerol concentration in the anolyte influences GOR rates under varying thermal conditions. This exploration is particularly important given that the viscosity and mass transport properties of glycerol solutions are highly sensitive to temperature [46, 47], and therefore, as temperature increases, the optimal glycerol concentration in the anolyte for maximum catalytic performance may shift. Figure 3B summarizes the GOR current density at the peak potential of CVs recorded for Pt/NiF in 3 M KOH with varying glycerol concentrations (0.5–3 M) and temperature (25–65°C) (see Figure S15 for comparison of data).

As discussed previously, low glycerol concentrations in the anolyte may result in insufficient reactant delivery to the catalyst surface, whereas excessive concentrations can lead to mass transport limitations and hinder the removal of oxidation intermediates due to increasing viscosity. Accordingly, GOR current densities decline when the concentration of glycerol in the anolyte exceeds 1.6 M, regardless of temperature. Figure S15 also reveals a clear trend: as temperature increases, the glycerol concentration in the anolyte at which the maximum current density of the Pt/NiF electrode is observed increases. This shift is likely due to the improved mass transport within the electrolyte, as well as the accelerated reaction kinetics at elevated temperatures due to enhanced $\ast\text{OH}$ adsorption on the catalyst surface, making more $\text{Pt}-\text{OH}$ active sites available for GOR [42, 48]. Furthermore, facilitated intermediate oxidation at high temperatures reduces catalyst poisoning but demands

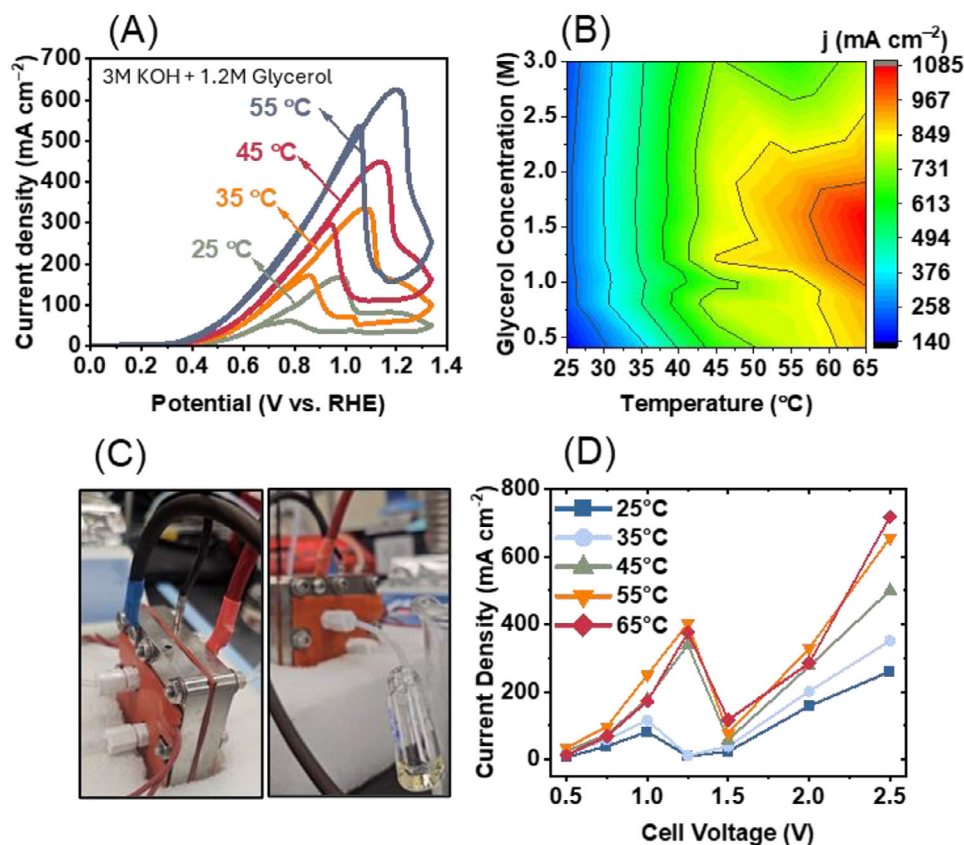


FIGURE 3 | Influence of temperature and glycerol concentration on electrochemical performance and product analysis in MEA configuration. (A) Cyclic voltammetry (CV) profiles of the Pt/NiF electrode at various temperatures in 3 M KOH + 1.2 M glycerol, recorded at a scan rate of 10 mV s⁻¹. (B) Heatmap showing the variation of maximum current density as a function of temperature and glycerol concentration, obtained from CV measurements at 10 mV s⁻¹. (C) Photographs of the MEA setup equipped with heating pads for temperature control and real-time collection of GOR output solution for subsequent product analysis. (D) Polarization curves of the MEA recorded during 5 min chronoamperometry (non-iR-corrected voltage) at different cell voltages and temperatures (the current density shows the average of the last 2 min of the operation).

a steady glycerol supply to maintain the activity. Therefore, at elevated temperatures, increasing the glycerol concentration becomes necessary to ensure efficient mass transport, continuous reactant delivery, and optimal GOR performance. Additional details on the temperature-dependent LSV profiles for Pt/NiF (Figure S16) and NiF (Figure S17), including the kinetics of GOR on Pt and Ni sites, Tafel slope analyses, and the interplay between GOR and OER at elevated potentials, are provided in the (Section S3 and Figures S18–S20).

Shifting focus from electrode-level kinetics to full MEA performance, Figure S21 presents 5 min chronoamperometry measurements of the Pt/NiF electrode in an MEA at different cell voltages and temperatures. As seen in the results collected at 25 and 35°C, in the GOR zone (<1.5 V), the current density reaches a maximum of 87 mA cm⁻² (1.0 V, 25°C) and 107 mA cm⁻² (1.0 V, 35°C). At higher cell voltages (1.25 and 1.5 V), a sharper decline in current density to ~13 and 21 mA cm⁻², respectively, is observed, confirming Pt deactivation, likely due to the formation of Pt-(hydr)oxide species as discussed previously. However, when the temperature increases beyond 45°C, the trend changes noticeably. At 1.25 V (45°C), the current density rises to 356 mA cm⁻² and remains stable throughout the duration of the test, resulting in a yellow solution of GOR product directly out of MEA. (Figure 3C). Similar behavior is observed at 55 and

65°C, with GOR current densities reaching 389 and 410 mA cm⁻² at 1.25 V, respectively. This sustained, high-rate performance at elevated temperatures is particularly notable, as it demonstrates that Pt-based catalysts can sustain significant GOR activity under thermally enhanced conditions, a behavior not widely reported in prior literature. Therefore, the results summarized in the polarization curve shown in Figure 3D, indicate two important conclusions: i) Increasing the MEA operation temperature significantly enhances both the GOR and OER performance, enabling current densities up to 410 mA cm⁻² (GOR zone) and 738 mA cm⁻² (GOR/OER zone), ii) Despite the temperature effect, catalyst deactivation due to Pt-(hydr)oxide formation at 1.5 V appears to be largely temperature-independent, indicating a fundamental limitation in terms of preserving catalytically active Pt surfaces under strongly oxidizing conditions. These findings highlight the critical role of operational parameters, including anolyte/catholyte composition, flow rate, and especially temperature, in tuning and optimizing the performance of glycerol-fed electrochemical systems

2.4 | Product Selectivity and Reaction Mechanism

To gain deeper insight into the mechanisms of glycerol conversion and product distributions during the GOR, we analyzed

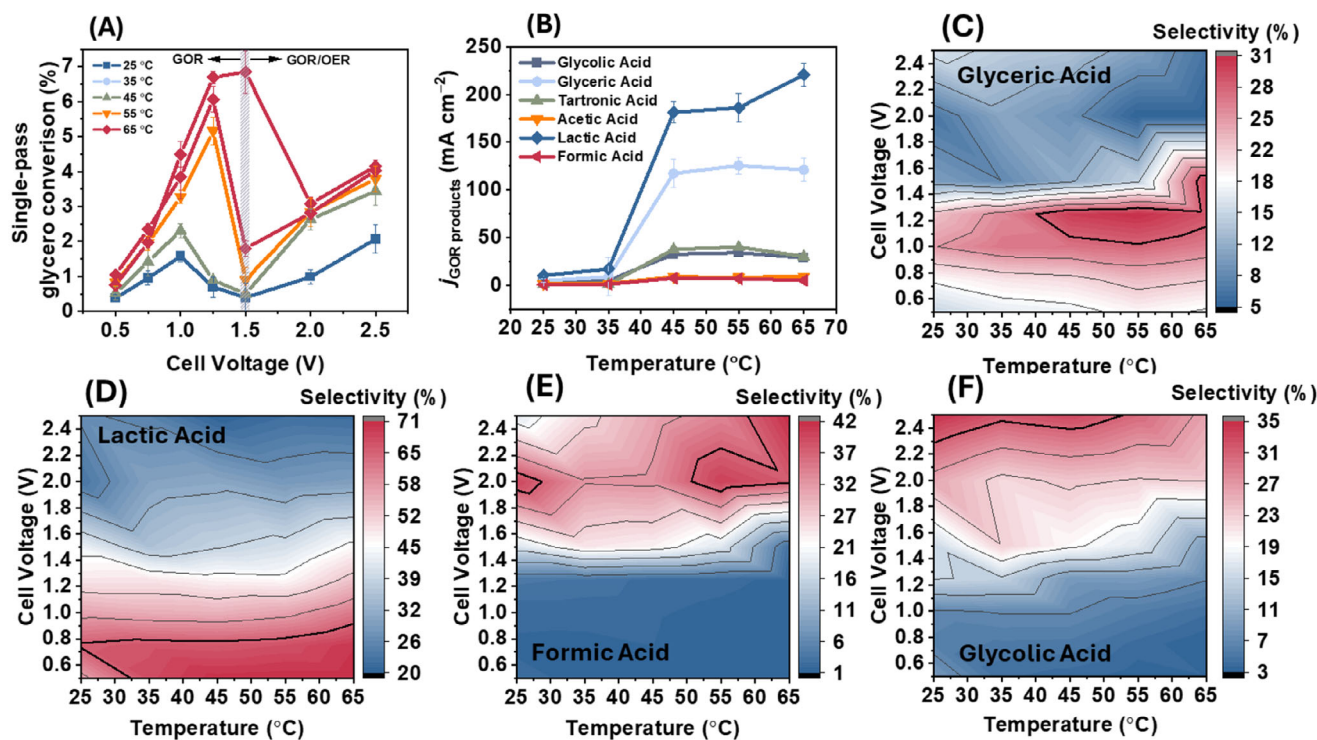


FIGURE 4 | Effect of temperature and cell voltage on glycerol conversion efficiency and product selectivity. (A) Single-pass glycerol conversion efficiency as a function of temperature and cell voltage in both the GOR and GOR/OER operational zones. (B) Variation in partial current densities of individual GOR products as a function of temperature at a fixed cell voltage. (C–F) Selectivity distributions of key liquid-phase products as functions of temperature and cell voltage: (C) glyceric acid, (D) lactic acid, (E) formic acid, and (F) glycolic acid. All selectivity distributions are obtained via single-pass glycerol conversion processes ($n=3$).

the liquid-phase products (Figure 3D) collected after 5 min chronoamperometry tests on the Pt/NiF electrode in the MEA. Figure 4A presents the single-pass glycerol conversion efficiency as a function of both MEA temperature and applied voltage. At 25°C and 35°C, the glycerol conversion at 1.25 V (GOR-zone) were 1.3% and 2.4%, respectively. In contrast, the glycerol conversion at 2.5 V (GOR/OER zone) reached approximately 2.1% and 3.3% at 25 and 35°C, respectively. These results suggest that the OER performance of the NiF substrate may assist glycerol oxidation by generating reactive *OOH intermediates on the catalyst surface. However, a clearly different trend emerges at elevated temperatures. When operating above 45°C, the highest glycerol conversion values shift to the GOR-dominant region (cell voltage < 1.5 V), reaching 5.1% at 45°C, 6.0% at 55°C, and 6.7% at 65°C. These values significantly surpass the conversion rates achieved in the GOR/OER regime at the same temperatures. This temperature-dependent behavior highlights a critical insight: operating at elevated temperatures enables the GOR to proceed more efficiently in the absence of OER, thereby overcoming the conversion limitations typically observed under ambient conditions and enhancing the overall efficiency of the electrochemical glycerol oxidation process.

The partial current densities achieved for the Pt/NiF electrode within the MEA toward producing glyceric acid, lactic acid, tartronic acid, glycolic acid, acetic acid, and formic acid, as a function of temperature at 1.25 V, are presented in Figure 4B. The $j_{\text{lactic acid}}$ increases dramatically from 34 mA cm^{-2} at 35°C to approximately 221 mA cm^{-2} at 65°C. A similar trend is observed

for glyceric acid, with a partial current density rising more than six-fold from 17 mA cm^{-2} at 25°C to around 120 mA cm^{-2} at 45°C. However, further increases in temperature result in a gradual decline in glyceric acid formation. The partial current densities achieved for the Pt/NiF electrode within the MEA toward producing glyceric acid, lactic acid, tartronic acid, glycolic acid, acetic acid, and formic acid as a function of temperature at 1.25 V are presented in Figure 4B. The partial current density of lactic acid increases dramatically from 34 mA cm^{-2} at 35°C to approximately 221 mA cm^{-2} at 65°C. A similar trend is observed for glyceric acid, which increases from 17 mA cm^{-2} at 25°C to $\sim 120 \text{ mA cm}^{-2}$ at 45°C, followed by a gradual decline at higher temperatures. These trends suggest a temperature-dependent shift in reaction selectivity from electrochemical formation of glyceric acid toward lactic acid, potentially driven by enhanced diffusion of intermediates and thermally activated non-Faradaic rearrangements. A more detailed mechanistic analysis of GOR on Pt, including the role of glyceraldehyde intermediates, Cannizzaro-type chemistry, and its impact on Faradaic efficiency, is provided in the Supplementary Information (Section S4 and Figures S22–S24).

Increasing the cell voltage to 2.5 V, where GOR and OER occur concurrently, leads to different product distribution trends. As shown in Figure S25, the first key observation is the lower partial current densities for both glyceric acid and lactic acid at 2.5 V compared to those observed at 1.25 V. For instance, a partial current density toward glyceric acid ($j_{\text{glyceric acid}}$) below 20 mA cm^{-2} is observed across all tested temperatures, possibly due to electrochemical oxidization either via C–C cleavage

into glycolic acid (C_2) and formic acid (C_1), or through an additional alcohol-to-carboxyl conversion to form tartronic acid (C_3) as suggested previously [26]. Figure 4B reveals a substantial increase in $j_{\text{formic acid}}$ with temperature, from 23 mA cm^{-2} at 25°C to 118 mA cm^{-2} at 65°C, implying that glyceric acid is predominantly converted via the C–C cleavage route to glycolic acid and formic acid, a finding that has been reported previously on Pt-based catalysts [49]. This conclusion is further supported by the concurrent rise in $j_{\text{glycolic acid}}$ (to 41 mA cm^{-2} at 55°C) and the consistently low $j_{\text{tartronic acid}}$ (<5 mA cm^{-2}), indicating that C–C bond cleavage is favored over additional oxidation at the secondary alcohol site. These trends emphasize the role of Ni active sites in facilitating C–C cleavage, particularly under GOR/OER conditions, an observation reported before [50]. On the other side, the j_{OER} increases considerably from about 171 mA cm^{-2} at 25°C to 524 mA cm^{-2} at 65°C when the cell voltage is 2.5 V. This observation corresponds to the total Faradaic efficiency values of GOR being less than 40% at any temperature, indicating that OER is the dominant Faradaic reactions at 2.5 V (see Figures S23–S25 for the assumption regarding the OER Faradaic efficiency). However, it should be mentioned that a portion of remaining Faradaic efficiency could be due to the presence of some GOR products remaining undetected by H-NMR.

A clearer comparison of product distribution is provided in Figure 4D–F, which depicts the selectivity trends for glyceric acid, lactic acid, glycolic acid, and formic acid, respectively. It is important to note that selectivity reflects the relative distribution of products, not the absolute conversion or Faradaic efficiency, therefore, the graph of concentrations of products (Figures S26–S29) should be evaluated simultaneously. As shown in Figure 4D, glyceric acid selectivity declines at cell voltages above 1.4 V, likely due to increased C–C cleavage, forming deeper oxidation products. At the other extreme, selectivity also diminishes below 0.8 V due to sluggish catalytic activity of Pt. Thus, optimal glyceric acid selectivity is achieved within a moderate voltage window of 1.2–1.4 V, especially at elevated temperatures. Lactic acid selectivity (Figure 4E) follows a similar trend, peaking at 71% under 1.3 V and 65°C. At voltages above 2.0 V, the system increasingly favors electrochemical pathways over chemical conversions, resulting in lactic acid selectivity dropping below 20% across all temperatures. This observation supports the understanding that lactic acid originates mainly from the chemical Cannizzaro-type rearrangement of glyceraldehyde, which is suppressed under more oxidizing conditions (see Figure S30A–E for the bar graphs of the selectivity distribution).

In contrast, the selectivity trends for glycolic acid and formic acid (Figure 4E,F) show an inverse relationship compared to C_3 products. Both products display enhanced selectivity at higher voltages, consistent with their origin as secondary products of glyceric acid oxidation. However, while formic acid selectivity continues to increase with temperature and voltage, glycolic acid selectivity begins to decline beyond 2 V. This divergence aligns well with the $j_{\text{glycolic acid}}$ trends shown in Figure S25, indicating that glycolic acid may be further oxidized or undergo decomposition under strongly oxidizing conditions, whereas formic acid remains stable and accumulates as the terminal product.

In addition to the anodic value-added chemicals, H_2 is also generated at the cathode. Figure 5A presents the amount of

H_2 produced (in μmol) as a function of temperature at 1.25 V (GOR zone). At 1.25 V, the system simultaneously delivers over 600 μmol of H_2 and a peak yield of 176 μmol of C_3 chemicals (including lactic acid, glyceric acid, and tartronic acid) when the temperature is 65°C. This concurrent generation of value-added products at both electrodes represents an optimal operational regime for maximizing system-wide economic value. In contrast, at 2.5 V, while H_2 evolution exceeds 1000 μmol , the anodic product distribution shifts dramatically, only ~ 50 μmol of formic acid is detected, with negligible quantities of C_3 products (Figure 5B). This disparity highlights that under high-voltage conditions, a significant fraction of the anodic current is consumed by OER, rather than GOR, supported by the Faradaic efficiency graphs shown in Figure S23 (see Figure S31 for a detailed voltage-dependent H_2 production profile as compared to C_3 chemical production).

Since the major bottleneck in the long-term glycerol electrooxidation is the inability of Pt-based electrodes to maintain performance beyond a few minutes primarily, which arises from the accumulation of intermediate species or passivating species on the electrode surface and mass transport limitations, it is important to assess the long-term performance of the proposed MEA system. For this purpose, we first conducted 24 h chronopotentiometry experiment at a constant current density of 300 mA cm^{-2} and a temperature of 65°C, with product samples collected at 1, 3, 6, and 24 h (see Tables S1 and S2 for Faradaic efficiency values) To mitigate voltage drift over time, the power supply was programmed to implement a 10 s off-cycle whenever the cell voltage exceeded 1.25 V (see the schematic of operation shown in Figure S32. These brief pauses (down to 0 V) may also assist in electrochemical surface recovery by facilitating the reduction of PtO_x and bound intermediates. As shown in Figure 5C, the system consistently maintained 300 mA cm^{-2} at an average operating voltage of 1.15 V, achieving a stable single-pass glycerol conversion of approximately 3.5% over 24 h (see Figure S33 for the oxidation times in each on-cycle). Importantly, the anodic product distribution remained largely unchanged throughout the test period, with cumulative C_3 selectivity remaining near 89% and lactic acid selectivity consistently around 50% (Figure S34). These results confirm the robustness of the proposed MEA configuration for prolonged and industrial-relevant operation of GOR.

To evaluate performance under higher current densities, we further tested the MEA at 500 mA cm^{-2} and 65°C for 24 h. As shown in Figure 5D, the system can deliver a current density of 500 mA cm^{-2} at just 1.2 V, by maintaining the average on-cycles of ~ 2 min (Figure S35). Such low cell voltage (1.2 V, still in GOR zone) at 500 mA cm^{-2} enables us to produce >175 mmoles of H_2 at the cathode along with 45 mmoles of C_3 chemicals at the anode (Figure 5E). Our proposed dynamic, self-regulating mode of operation (that is different from conventionally named pulsed electrolysis) enables high-performance GOR performance below ~ 1.25 V, achieving a record-high single-pass conversion of nearly 9% for a Pt-based electrode, outperforming the best reported MEA current densities in alkaline GOR systems to date (Figure 5F, also see Table S3 for the comprehensive comparison of our work with existing literature on glycerol-fed MEA coupled with HER). When operated under continuous anolyte recirculation, however, the chronopotentiometry measurements show a gradual increase

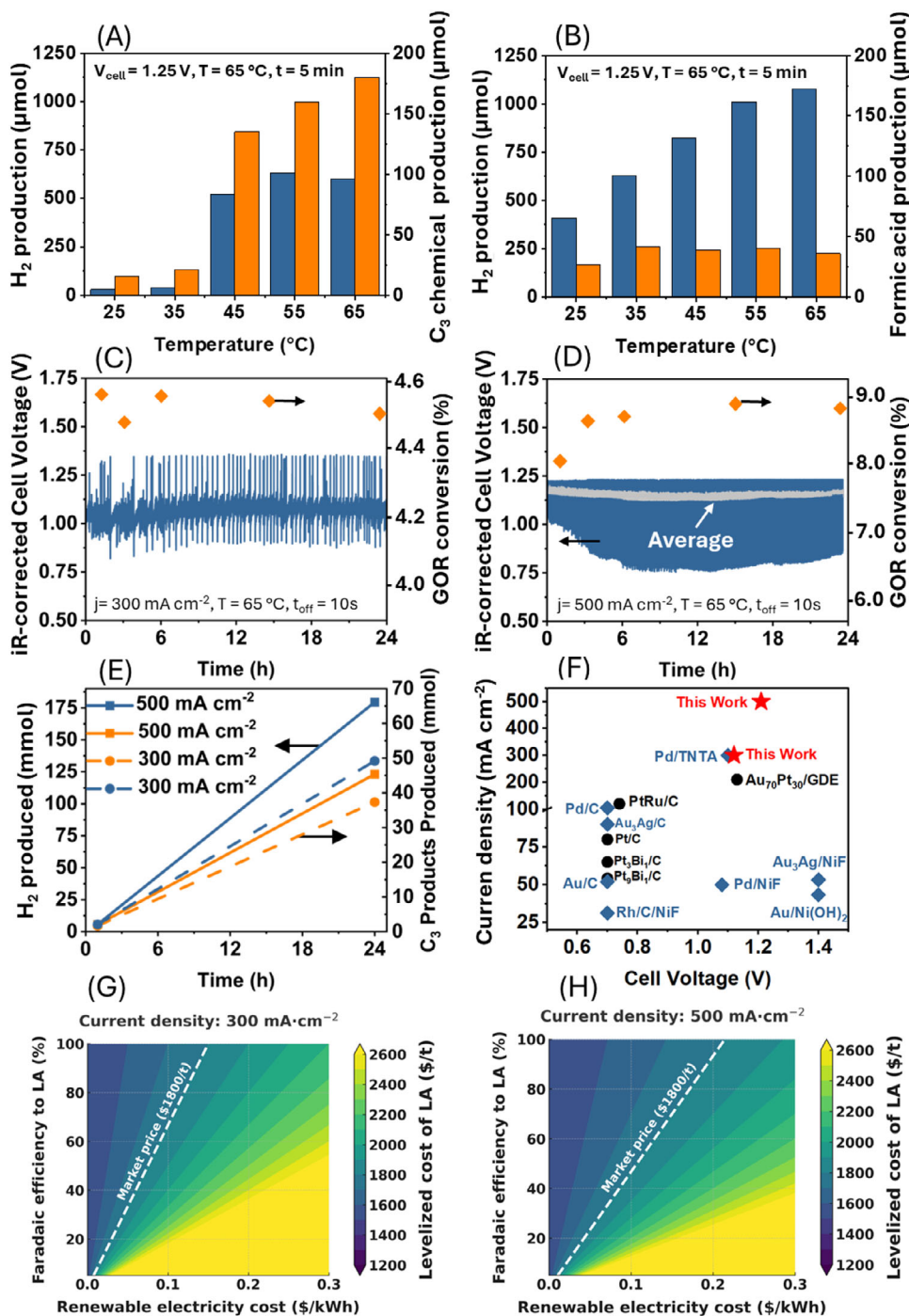


FIGURE 5 | Hydrogen co-production, voltage-regulated operation, and performance benchmarking of the MEA system. Comparison of H_2 production (in μmol) with: (A) C_3 chemical production at 1.25 V and (B) formic acid production at 2.5 V during 5 min chronoamperometry experiments. (C) Chronopotentiometry profile of MEA operation for a single-pass glycerol conversion at a constant current density of 300 mA cm^{-2} , incorporating a programmed 10 s off-cycle whenever the cell voltage exceeded 1.25 V to maintain long-term performance. (D) Chronopotentiometry profile of MEA operation for single-pass glycerol conversion at a constant current density of 500 mA cm^{-2} , incorporating a programmed 10 s off-cycle whenever the cell voltage exceeded 1.45 V. (E) Total H_2 and C_3 chemical production as function of operation time at current densities of 300 and 500 mA cm^{-2} . (F) Comparison of average current densities achieved by the optimized MEA system in this study with those reported for other state-of-the-art Pt-containing catalysts in the literature (black circles indicate the Pt-containing catalyst, and blue squares indicate the other catalysts). (G,H) Levelized cost of Lactic acid electrosynthesis as a function of Faradaic efficiency and renewable energy cost.

in cell voltage, reaching ~ 1.45 V after 24 h (Figure S36). This deviation is attributed to progressive glycerol depletion and possible reoxidation of intermediate GOR products, underscoring the need to tailor feed concentration and flow conditions for closed-loop operation.

As shown in Figure S37A, constant operation results in a rapid voltage rise, exceeding 2.0 V within ~ 30 min and shifting the system into an OER-dominated regime. Under pulsed operation (Figure S37B), the initial low-voltage behavior is maintained only for the first few hours. However, by ~ 4 h, the cell similarly stabilizes above 2.0 V. In contrast, the dynamic self-regulating mode successfully sustains an average cell voltage of 1.25 V for over 24 h at 500 mA cm^{-2} , well below the OER threshold and firmly within the GOR-dominant region. Consistent with these profiles, the product Faradaic efficiencies (Figure S37C) show that dynamic operation delivers significantly higher C_3 selectivity at elevated glycerol conversion, whereas both constant and pulsed modes, despite producing more H_2 , strongly suppress C_3 formation (Figure S38).

We also examined morphological changes in the electrodes after 24 h of electrolysis at 500 mA cm^{-2} to evaluate their durability. The SEM images of used electrodes at different magnifications (Figure S39A–D) and their EDX mapping (Figure S39E–J) still reveal the presence of Pt nano-dendritic structures on the surface of NiF, even after 24 h of operation under a current density of 500 mA cm^{-2} and strongly alkaline environment. However, closer inspection reveals localized regions with reduced Pt coverage, suggesting partial mechanical detachment and/or electrochemical leaching of Pt during the reactions. This observation is supported by ICP-OES results (Figure S40) revealing that during the first 6 h of operation, the total Pt loading in the Pt/NiF electrode drops from 58.4 to $51.6 \mu\text{g cm}^{-2}$. However, no meaningful change in Pt loading is observed during the next 18 h, indicating that most of the Pt loss occurs early in the reaction. These results correspond to a Pt mass loss rate of approximately $1.12 \mu\text{g h}^{-1}$ during the first 6 h, followed by a much slower rate of $\sim 0.05 \mu\text{g h}^{-1}$ thereafter. To rule out bulk electrolyte effects on the stability of Pt, we measured the pH of both inlet and outlet streams. As shown in Figure S41, the single-pass conversion induces only a negligible change in the pH ($\Delta\text{pH} < 0.1$), confirming that loss of Pt cannot be attributed to variations in electrolyte pH.

X-ray diffraction (XRD) was used to examine the phase structure of Pt/NiF before and after electrolysis (Figure S42A). Compared with the fcc Pt (PDF#04-0802) and Ni (PDF#04-0850) references, the as-prepared Pt/NiF shows an upshift in the diffraction angle (2θ) for the Pt reflections and a downshift in 2θ for the Ni reflections (Figure S42B). This reciprocal shift indicates the formation of a Pt–Ni alloy, in which partial Pt incorporation into Ni expands the Ni lattice, while Ni incorporation into Pt slightly contracts the Pt lattice, consistent with Vegard-type behavior. After pulsed electrolysis, the Pt peaks disappear within the signal-to-noise limits of the scan, suggesting a pronounced reduction in crystalline Pt content at the probed depth, possibly due to the loss of Pt. In contrast, the post-dynamic-electrolysis sample retains distinct Pt peaks that are narrower (indicating improved crystallinity or decreased microstrain) and further right-shifted relative to the fresh electrode, implying enhanced alloying and increased Ni incorporation into the Pt-rich phase formed during

operation (Figure S42C). For Ni, dynamic electrolysis causes the main Ni peak to shift toward the reference Ni position. This shift could be due to the change in the surface Ni species that locally contracts the lattice.

These XRD observations are further supported by X-ray photoelectron spectroscopy (XPS) analysis (Figure S43). The high-resolution Pt 4f spectra of the fresh (Figure S43A) and post-electrolysis (Figure S43B) electrodes reveal both metallic Pt^0 and oxidized Pt^{2+} (Pt–O) components. The metallic Pt^0 peak shifts from 71.2 eV in the fresh sample to 70.6 eV after electrolysis, consistent with stronger Pt–Ni electronic interaction (enhanced alloying) [36]. Comparison of the Ni 2p spectra (Figure S43C,D) provide two key insights: i) the fresh electrode exhibits a metallic Ni^0 feature near 852.7 eV, whereas the post-electrolysis sample lacks this signal within the probed depth, indicating oxidation to Ni–O species; and ii) the $\text{NiOOH}/\text{Ni}(\text{OH})_2$ ratio increases after electrolysis, suggesting that long-term operation promotes the conversion of near-surface $\text{Ni}(\text{OH})_2$ into NiOOH . The latter also supports the observed trend in the shift of the Ni peak in the XRD pattern, as NiOOH (Ni^{3+}) shows a smaller lattice spacing compared to $\text{Ni}(\text{OH})_2$, causing the lattice contraction [51].

We also performed a base-case technoeconomic analysis for co-production of lactic acid and glyceric acid to place our operating strategy in an economic context (see Section S5 and Figure S44; Tables S4–S9 for the details of technoeconomic analysis). At 300 mA cm^{-2} and 31 % Faradaic efficiency (Figure 5G), the calculated levelized cost of lactic acid is approximately $\$1786 \text{ t}^{-1}$, while at 500 mA cm^{-2} (Figure 5H) and 40 % efficiency, the levelized cost decreases to $\$1671 \text{ t}^{-1}$, indicating a lower production cost per tonne due to reduced energy intensity at higher current density. However, the total estimated profit from the techno-economic analysis is $\$499 \text{ t}^{-1}$ for the 500 mA cm^{-2} case while for the 300 mA cm^{-2} one is $\$612 \text{ t}^{-1}$. This discrepancy arises because, despite a lower levelized cost, the 500 mA cm^{-2} system operates about 3 h less (due to 1500 OFF cycles while 300 mA cm^{-2} has only 50 off cycles), which reduces total plant revenue.

While dynamic operation provides clear advantages in maintaining low-voltage and performance that is selective toward the GOR over extended operation, it also introduces operational trade-offs. The periodic 10 s relaxation steps reduce the effective duty cycle, and therefore the time-averaged productivity, and such intermittent operation may not be desirable for some industrial processes that require uninterrupted power delivery. These observations align with broader trends in the emerging pulsed-electrolysis literature, where improvements in stability or selectivity often come at the cost of reduced uptime or added operational complexity [29]. Accordingly, we view dynamic operation not as a universal replacement for continuous electrolysis, but as a practical mitigation strategy under the high-current, single-pass conditions employed here, conditions under which continuous operation rapidly drifts into high-voltage OER-dominant behavior.

Looking ahead to scale-up, several factors must be considered for implementing this highly concentrated electrolyte strategy (3 M KOH + 1 M glycerol). First, carbonation is a well-known issue in alkaline systems: exposure to ambient CO_2 promotes the formation of carbonate/bicarbonate species and, at high concentrations, K_2CO_3 precipitates, which can alter electrolyte

composition, increase ohmic losses, and foul flow channels over time. Second, the elevated viscosity and ionic strength of concentrated KOH/glycerol mixtures can hinder mass transport, necessitating optimized flow-field designs, more intensive pumping, or periodic electrolyte refresh to sustain performance at scale. Third, downstream product separation from a strongly alkaline matrix remains challenging. Neutralization with H_2SO_4 is one viable route, but it entails substantial acid consumption (an economic barrier) and generates K_2SO_4 , whose limited market size and relatively low value reduce the benefit of its recovery [21]. Alternative separation strategies (e.g., electrodialysis or reactive extraction) therefore represent important directions for future research. Consequently, although the leveled cost indicates improved cost efficiency per unit product under the conditions explored here, overall profitability in a technoeconomic sense will ultimately depend on how these carbonation, mass-transport, and separation challenges are managed under realistic operating conditions.

3 | Conclusion

This work establishes a high-performance membrane electrode assembly for glycerol electrooxidation, combining mechanistic insight with scalable reactor design. Potential-resolved EIS analysis reveals distinct operational regimes corresponding to hydroxide adsorption, selective glycerol oxidation, and oxygen evolution on Pt/Ni electrode, highlighting the role of Pt–OH species in tuning catalyst activity and selectivity. Leveraging these insights, we demonstrate 24 h operation of membrane electrode assembly at 500 mA cm^{-2} with an average cell voltage of just 1.21 V, outperforming all previously reported MEA systems for glycerol-fed electrolysis to date. The system achieves >88% selectivity toward C_3 products, with lactic acid accounting for ~50% of the product distribution (partial current density $\sim 227 \text{ mA cm}^{-2}$) while reaching ~9% single-pass glycerol conversion. In total, more than 175 mmol of H_2 and 45 mmol of C_3 products are co-generated in a single operational cycle (24 h). These findings offer both operational and mechanistic principles for long-term, low-voltage electro-conversion of biomass alcohols and provide a transferable strategy for coupling H_2 production with waste valorization in next-generation, energy-efficient electrolyzer platforms.

4 | Methods

4.1 | Materials

NiF was purchased from MTI Inc. Potassium hexachloroplatinate (K_2PtCl_6), perchloric acid (HClO_4), hydrochloric acid (HCl), acetone, isopropanol (IPA), KOH (98.0%), and glycerol (99.0%) were purchased from Sigma Aldrich. All chemicals were used fresh and without further purification.

4.2 | Catalyst Preparation

The Pt/NiF catalyst was synthesized with a slight modification to our previously published protocol [26]. Briefly, a $1 \text{ cm} \times 1 \text{ cm}$ piece of NiF was cut and sequentially ultrasonicated in acetone, IPA,

HCl (3 M), and Millipore water for 5 min each to clean the surface and then remove the native oxide layer on Ni. The NiF was then submerged in a solution consisting of 1 mL of 10mM K_2PtCl_6 in 8 mL of 0.1 M HClO_4 for 4 min, causing the electrode to become a dark gray color due to the galvanic deposition of Pt. The prepared samples were then rinsed with plenty of Millipore water (water resistivity $\sim 18.2 \text{ } \Omega \cdot \text{cm}$) and dried in an oven at 150°C for 1 h.

To investigate the role of Pt in the performance of Pt/NiF toward different anodic reactions, pure NiF was also prepared using the method above, without using the K_2PtCl_6 solution. Also, to compare the OER activity of the synthesized catalyst with a commercial benchmark catalyst, a commercial IrO_x electrode ($1 \text{ cm} \times 1 \text{ cm}$, Dioxide Material) was used. For the counter electrode (in three electrode H-cell) or cathode used in MEAs, a $1 \times 1 \text{ cm}^2$ commercial Pt/C catalyst ($0.5 \text{ mg}_{\text{Pt}}/\text{carbon}$, fuel cell store) was used to ensure consistency throughout the measurements.

4.3 | Membrane Electrode Assembly Configuration

The electrolyzer hardware used in this study was purchased from Dioxide Materials Inc., consisting of titanium and stainless-steel current collectors at the anode and cathode, respectively. To ensure proper integration of the NiF electrodes into a zero-gap assembly without compressing the NiF, the MEA was assembled using three silicone-based gaskets providing about 1 mm thickness each (see Figure S45A,B). The anion exchange membrane (Sustainion X37) was activated by immersing it in 1 M KOH for 24 h. Then a $2 \text{ cm} \times 2 \text{ cm}$ area was cut, rinsed thoroughly with Millipore water, and placed on the gasket. A Pt/C cathode is then placed on the top of the electrolyte membrane to prepare the MEA, and another gasket is placed around the Pt/C cathode to sandwich the electrolyte membrane. The electrolyzer hardware was then assembled and connected to two peristaltic pumps, flowing the electrolytes on both anodic and cathodic sites. The anolyte used in the experiments was 3 M KOH with varying glycerol concentration (depending on the experiment) and was passed through the MEA with the single pass method at a flow rate of 0.5 mL min^{-1} . The catholyte was a 4 M KOH solution that was fed into the MEA at a flow rate of 5 mL min^{-1} .

For the experiments requiring changes in temperature, two thermal heating pads (purchased from Dioxide Materials) were attached on both sides of the electrolyzer hardware (the orange pads shown in Figure S45C). The thermocouple was placed in a hole drilled into the anode size of the electrolyzer hardware, with an approximate distance of 30 mm from the external surface, while the temperature was controlled by a digital unit. To ensure that temperature gradients do not form within the electrolyzer during operation, both the anolyte and catholyte were also pre-heated to the desired temperature prior to use, while also being purged with Ar to remove any dissolved gases such as oxygen.

4.4 | Electrochemical Tests

Electrochemical impedance spectroscopy (EIS) was conducted prior to MEA testing to measure the solution and electrolyte membrane resistance, enabling manual iR-correction of the

recorded cell voltages. This correction was essential to ensure meaningful performance comparisons under varying operating conditions. For these measurements, both AC and DC amplitudes were set to 10 mV, and the frequency ranged from 1 to 100 mHz. Additionally, the Bode plots (change in the phase angle by applied frequency) derived from EIS were employed to gain insight into charge distribution on the electrode surface and underlying mechanisms enabling GOR and OER. For these potential-dependent EIS, measurements were performed over a frequency range from 10^{-2} to 10^3 Hz at fixed DC amplitude (10 mV), while the AC potential varied from 0 to 1.6 V versus RHE in 0.05 V increments. Chronoamperometry tests were used to construct polarization curves and collect electrolyte samples for product analysis. For the H-cell measurements, the working, reference, and counter electrodes were Pt/NiF, Hg/HgO, and Pt/C, respectively. The electrode potential in the three-electrode experiments was measured relative to the reversible Hg/HgO redox couple and converted to the RHE scale using:

$$E_{\text{RHE}} = E_{\text{Hg/HgO}} + E_{\text{Hg/HgO}}^{\circ} + 0.0591 \times \text{pH} \quad (1)$$

With the assumption of constant pH during H-cell experiments, as CV/LSV/EIS produce <10 mM of products, which is insufficient to significantly alter the bulk electrolyte pH.

4.5 | Product Analysis

Glycerol oxidation products were collected directly from the MEA outlet after each chronoamperometry experiment and stored in the refrigerator prior to further analyses. Product identification and quantification were performed using proton nuclear magnetic resonance ($^1\text{H-NMR}$, Bruker 600 MHz spectrometer, McMaster University NMR Facility), following protocols described in our previous work [25]. For each analysis, 700 μL of the anolyte was combined with 35 μL of an internal standard solution, which contained 2.98 μL of dimethyl sulfoxide (DMSO) and 0.0988 g of phenol dissolved in 10 mL of deuterium oxide (D_2O).

To calculate the selectivity for the products of glycerol oxidation, Equation (2) was used:

$$\text{Selectivity \%} = \frac{n_{\text{desired product}}}{\sum n_{\text{all products}}} \times 100 \quad (2)$$

Moreover, the glycerol conversion was calculated using the following equation (Equation (3)):

$$\text{Glycerol conversion \%} = \frac{\sum n_{\text{all products}}}{(n_{\text{Glycerol, initial}})} \times 100 \quad (3)$$

To calculate the Faradaic Efficiency for each product, Equation (4) was used:

$$\text{Faradaic Efficiency \%} = \frac{z_i n_i F}{Q} \times 100 \quad (4)$$

where n_i is the molar concentration of the product, z_i is the number of electrons needed to generate that product directly from glycerol, and F is the Faradaic constant (96845 C mol^{-1}). Q

is also the total input charge to the system during electrolysis. When evaluating Faradaic efficiency in glycerol electrooxidation, two critical considerations must be considered. First, the number of electrons theoretically required to produce a given product “electrochemically” may differ from the actual number of electrons transferred in the reaction. This discrepancy arises due to chemical transformations, such as rearrangements or condensations, that occur among intermediate or final products in the reaction mixture. As a result, the calculated Faradaic efficiency values can be difficult to interpret and may not reflect purely electrochemical pathways. Second, in this study, we report apparent Faradaic efficiencies, where the total Faradaic efficiency of all detected products is normalized to 100%. This normalization facilitates direct, apple-to-apple comparison of product distributions across different experimental conditions, even though it does not account for undetected products or side reactions that may consume charge.

4.6 | Physical and Chemical Characterizations

The SEM images were captured using the FEI Magellan 400 at the Canadian Center for Electron Microscopy (CCEM) at 5 KV and secondary electron mode. Or inductively coupled plasma optical emission spectroscopy (ICP-OES) analysis using an Agilent 5900 ICP-OES, 100 mL of 6 M nitric acid (HNO_3) was heated to 80°C . Subsequently, 1 cm^2 (geometric surface area) of the electrodes were placed in this solution for 72 h. Then, 1 mL of the dissolved samples was diluted by a factor of 100 before being used for ICP-OES measurements.

Acknowledgements

The authors acknowledge financial support from the Natural Sciences and Engineering Research Council of Canada (NSERC) Alliance program, the Mitacs Accelerate program, and Nutrien Ag Solutions Inc. (Canada). Product analyses were conducted at the McMaster Nuclear Magnetic Resonance (NMR) Facility. Scanning electron microscopy was performed at the Canadian Centre for Electron Microscopy (CCEM). X-ray diffraction was conducted at the McMaster Analytical X-ray Diffraction Facility (MAX). X-ray photoelectron spectroscopy (XPS) was conducted at Surface Science Western (SSW) at Western University.

Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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